AP/ITEC 4220 A - Modern Approaches to Data Management: Database Management Systems

Milestone 3

Group 2

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12 queries in XML Document

1. List all accounts with their customers details (account ID, guest flag, and customer name, address, email, and phone) customer_accounts.xml

Accounts and their customers

```
select xmlroot(xmlelement("CustomerAccounts",
      xmlagg(
            xmlelement("Account", xmlattributes(a.account_id AS "accountId",
            a.guestLogin AS "guestFlag"),
                  xmlelement("Customer", xmlattributes(
                  a.customer_ref.customer_id AS "customerId"),
                  xmlforest(
                     a.customer_ref.name AS "Name",
                     a.customer_ref.address AS "Address",
                     a.customer_ref.email AS "Email",
                     a.customer_ref.phoneNum AS "Phone"))))),
       VERSION '1.0') AS document from account a;
```

2. For every account ID, show customer details and all related orders, including order ID, order date, and total amount. customer_orders.xml

Orders with account and customer details

```
select xmlroot(xmlelement("CustomerAccounts",
          xmlagg(
             xmlelement("Account", xmlattributes(a.account_id AS "accountId"),
                xmlelement("Customer",
                   xmlattributes(a.customer_ref.customer_id AS "customerId"),
                   xmlforest(
                      a.customer_ref.name AS "Name",
                      a.customer_ref.address AS "Address",
                      a.customer_ref.email AS "Email",
                      a.customer_ref.phoneNum AS "Phone")),
                (select xmlagg(
                           xmlelement("Order",
                             xmlattributes(o.accountorder_id AS "orderId"),
                             xmlforest(
                                o.order_date AS "OrderDate",
                                o.total_amount AS "TotalAmount")))
                   from accountorder o
                   where o.account_ref = REF(a))
```

))), VERSION '1.0') AS document from account a;

3. For cart lines that contain videogames, show the customer, order, and videogame information game_cartlines.xml

Videogame cart lines per order

```
select xmlroot(xmlelement("VideogameCartLines",
xmlagg( xmlelement("CartLine", xmlattributes(cl.cartline_id as "cartlineId",  cl.quantity as "Quantity"),
xmlelement("Order", xmlattributes(o.accountorder_id as "orderId", o.order_date  as "OrderDate"), xmlelement("OrderTotal", o.total_amount),
xmlelement("Customer", xmlattributes(c.customer_id as "customerId"), xmlelement("Name", c.name))),
xmlelement("Videogame", xmlattributes( vg.product_id AS "GameId", vg.platform AS "Platform", vg.estb_rating AS "Rating"), xmlforest(vg.name as "Title", vg.base_price as "BasePrice"), xmlelement("UnitPriceCharged", cl.unit_price))))), version '1.0') as document
from cartline  cl, accountorder o, account  a, customer  c, videogame vg
where cl.order_ref = ref(o) and o.account_ref  = ref(a) and a.customer_ref = ref(c)  and cl.product_ref = ref(vg);
```

4. For all cart lines that contain accessories, show the customer, order, accessory information accessory_cartlines.xml

Accessory cart lines per order

```
select xmlroot(xmlelement("AccessoryCartLines",
xmlagg( xmlelement("CartLine", xmlattributes(cl.cartline_id  as "cartlineId",  cl.quantity  as "Quantity"),
xmlelement("Order", xmlattributes(o.accountorder_id as "orderId", o.order_date  as "OrderDate"), xmlelement("OrderTotal", o.total_amount),
xmlelement("Customer", xmlattributes(c.customer_id as "customerId"), xmlelement("Name", c.name))),
 xmlelement("Accessory", xmlattributes(ac.product_id as "AccessoryId", ac.accessory_type as "Type", ac.color  as "Color"),
xmlforest(ac.name as "Name", ac.base_price as "BasePrice"),
xmlelement("UnitPriceCharged", cl.unit_price))))),
version '1.0') as document from cartline  cl, accountorder o, account  a, customer  c,
accessories  ac
```

```
where cl.order_ref = ref(o) and o.account_ref  = ref(a)  and a.customer_ref = ref(c) and
cl.product_ref = ref(ac);
```

5. List every supplier information and contracts including contract ID, terms , price, expiry date, and supplier rating. suppliers.xml

Suppliers and their supply contracts

```
select xmlroot(xmlelement("Suppliers",
        xmlagg(
           xmlelement("Supplier",
              xmlattributes(s.supplier_id AS "supplierId", s.rating AS "rating"),
        xmlforest(s.name AS "Name", s.address AS "Address", s.phone AS  "Phone"),
              xmlelement("Contracts",
                 (select xmlagg(
                           xmlelement("Contract",
                             xmlattributes(sc.supplycontract_id AS "contractId"),
                              xmlforest(sc.terms AS "Terms", sc.cost_price AS "CostPrice",
                           sc.expiry_date AS "ExpiryDate")))
                    from supplycontract sc
                    where sc.supplier_ref.supplier_id = s.supplier_id)
    ))))), VERSION '1.0') AS document from supplier s;
```

6. For every supplier, list all videogames they supply, including product ID, name, price, platform, rating, and contract price. game_supplier.xml

Suppliers and their videogames

```
select xmlroot(xmlelement("SupplierVideogames",
        xmlagg(
           xmlelement("Supplier",
              xmlattributes(s.supplier_id AS "supplierId"),
              xmlelement("Name", s.name),
              xmlelement("Videogames",
                 (select xmlagg(
                           xmlelement("Game",
                             xmlattributes(vg.product_id AS "gameId"),
                             xmlforest(
                                vg.name AS "Title",
```

```
                    vg.base_price AS "BasePrice",
                    vg.platform AS "Platform",
                    vg.estb_rating AS "Rating",
                    sc.cost_price AS "ContractCost")))
                from supplycontract sc, videogame vg
                where sc.supplier_ref.supplier_id = s.supplier_id AND
            vg.supplycontract_ref.supplycontract_id = sc.supplycontract_id)
)))), VERSION '1.0') AS document from supplier s;
```

7. For every supplier, list all accessories they supply, including product ID, name, price, type, colour, and contract price. accessory_supplier.xml

Suppliers and their accessories

```
select xmlroot(xmlelement("SupplierAccessories",
xmlagg( xmlelement("Supplier", xmlattributes(s.supplier_id as "supplierId"),
xmlelement("Name", s.name), xmlelement("Accessories", (select xmlagg(
xmlelement("Accessory", xmlattributes(ac.product_id as "accessoryId"),
xmlforest( ac.name as "Name", ac.base_price as "BasePrice", ac.accessory_type as "Type",
ac.color as "Color", sc.cost_price as "ContractCost" )))
from supplycontract sc, accessories ac
where sc.supplier_ref = ref(s) and ac.supplycontract_ref = ref(sc))))),
version '1.0') as document from supplier s;
```

8. For each customer, compute and show the total amount they have spent and the number of orders they placed. order_complete.xml

Orders with all cart lines (videogames + accessories)

```
select xmlroot(xmlelement("OrdersWithCartLines",
xmlagg(xmlelement("Order", xmlattributes(o.accountorder_id as "orderId"),
xmlelement("OrderInfo", xmlforest(o.order_date as "OrderDate", o.total_amount as
"TotalAmount" ), xmlelement("Customer", xmlattributes(c.customer_id as "customerId"),
xmlelement("Name", c.name))), xmlelement("CartLines",
(select xmlagg(x) from (
-- videogame lines
select xmlelement("CartLine", xmlattributes(clv.cartline_id as "cartlineId", 'Videogame' as
"productType"), xmlforest( clv.quantity as "Quantity", clv.unit_price as "UnitPrice"),
xmlelement("Videogame", xmlforest( vg.product_id   as "ProductId", vg.name as "Name",
```

```
vg.base_price as "BasePrice", vg.platform as "Platform", vg.estb_rating  as "Rating")) ) as x
from cartline clv, videogame vg
where clv.order_ref = ref(o) and clv.product_ref = ref(vg)
union all
-- accessory lines
select xmlelement("CartLine", xmlattributes(cla.cartline_id as "cartlineId", 'Accessory' as
"productType"), xmlforest(cla.quantity as "Quantity",cla.unit_price as "UnitPrice"),
xmlelement("Accessory", xmlforest( ac.product_id as "ProductId", ac.name as "Name",
ac.base_price as "BasePrice", ac.accessory_type as "Type",ac.color as "Color" ))) as x
from cartline cla, accessories ac
where cla.order_ref   = ref(o) and cla.product_ref = ref(ac)))))),
version '1.0' ) as document
from accountorder o, account a, customer c
where o.account_ref  = ref(a) and a.customer_ref = ref(c);
```

9. For each customer, show number of purchases their account made (Group By) purchase.xml

number of orders per customer and total spent

```
select xmlroot(xmlelement("CustomerTotals",
        XMLAGG(XMLELEMENT("Customer",
            XMLATTRIBUTES(t.customer_id AS "customerId"),
                XMLFOREST(t.name AS "Name", t.order_count AS "OrderCount",
                t.total_spent AS "TotalSpent")))),VERSION '1.0') AS document
FROM (SELECT
        -- Select Customer attributes
        a.customer_ref.customer_id AS customer_id,
        a.customer_ref.name AS name,
        -- Aggregate Orders
         COUNT(o.accountorder_id) AS order_count,
         SUM(o.total_amount) AS total_spent
         FROM account a, accountorder o
         WHERE o.account_ref.account_id = a.account_id
         GROUP BY a.customer_ref.customer_id, a.customer_ref.name) t;
```

10. For each videogame platform (e.g., PS5, Switch, XBOX, Steam), show total quantity sold and total revenue. (Group By) platform_sales.xml

Sales per videogame platform

```
select xmlroot(xmlelement("PlatformSales",
```

```
xmlagg(xmlelement("Platform", xmlforest( t.platform as "PlatformName", t.total_qty as
"TotalQuantity", t.total_revenue as "TotalRevenue")))), version '1.0') as document
from (select vg.platform, sum(cl.quantity) as total_qty, sum(cl.quantity * cl.unit_price) as
total_revenue from cartline cl, accountorder o, videogame vg where cl.order_ref  = ref(o) and
cl.product_ref = ref(vg) group by vg.platform) t;
```

11. Show daily sales revenue generated (Group By) daily_sales.xml

Daily sales per order date

```
SELECT XMLROOT(XMLELEMENT("DailySales",
        XMLAGG(
          XMLELEMENT("Day", XMLFOREST(
              t.order_date AS "OrderDate",
              t.order_count AS "OrderCount",
              t.daily_total AS "TotalAmount")
           ))), VERSION '1.0') AS document
FROM (SELECT
            o.order_date,
            COUNT(o.accountorder_id) AS order_count,
            SUM(o.total_amount) AS daily_total
          FROM accountorder o
          GROUP BY o.order_date) t;
```

12. For each supplier, show the total revenue generated from all videogames and accessories supplied by them (Group By) supplier_revenue.xml

Total revenue per supplier (games + accessories)

```
select xmlroot(xmlelement("SupplierRevenue", xmlagg(xmlelement("Supplier",
xmlattributes(t.s_id as "supplierId"), xmlforest( t.s_name as "Name", t.total_revenue as
"TotalRevenue")))), version '1.0' ) as document
from (select s_id, s_name, sum(amount) as total_revenue from (
-- revenue from videogames
select s.supplier_id as s_id, s.name as s_name, cl.quantity * cl.unit_price as amount
from supplier s, supplycontract sc, videogame vg, cartline cl, accountorder o
where sc.supplier_ref= ref(s)and vg.supplycontract_ref = ref(sc) and cl.product_ref= ref(vg)
and cl.order_ref= ref(o)
union all
-- revenue from accessories
```

```
select s.supplier_id as s_id, s.name as s_name, cl.quantity * cl.unit_price as amount
from supplier s, supplycontract sc, accessories ac, cartline cl, accountorder o
where sc.supplier_ref = ref(s)  and ac.supplycontract_ref = ref(sc) and cl.product_ref =
ref(ac) and cl.order_ref = ref(o) ) group by s_id, s_name) t;
```

Oracle XSU

List all videogames where the contract cost is less than the public base price. Show the game ID, title, platform, base price, and the exact contract cost and supplier name.
deal.xml

```
OracleXML getXML -user "grp2/grp22here" -conn
"jdbc:oracle:thin:@sit.itec.yorku.ca:1521/studb10g" \
-rowsetTag "LowCostGames" -rowTag "GameDeal" \
"SELECT
   vg.product_id AS GameID,
   vg.name AS Title,
   vg.platform AS Platform,
   vg.base_price AS BasePrice,
   vg.supplycontract_ref.cost_price AS ContractCost,
   vg.supplycontract_ref.supplier_ref.name AS Supplier
FROM videogame vg
WHERE vg.supplycontract_ref.cost_price < vg.base_price"
```

List all accounts, including the customer's name and address, and include the total number of orders placed by that specific account account_summary.xml

```
OracleXML getXML -user "grp2/grp22here" -conn
"jdbc:oracle:thin:@sit.itec.yorku.ca:1521/studb10g" \
-rowsetTag "AccountSummary" -rowTag "AccountDetails" \
"SELECT
   a.account_id,
   a.customer_ref.name AS CustomerName,
   a.customer_ref.address AS CustomerAddress,
   (SELECT COUNT(*) FROM accountorder o WHERE o.account_ref = REF(a)) AS
TotalOrders
FROM account a"
```

List all accessories information and the calculated price using the override GetPrice() method. Include the name of the supplier responsible for the contract.
override_price.xml

```
OracleXML getXML -user "grp2/grp22here" -conn
"jdbc:oracle:thin:@sit.itec.yorku.ca:1521/studb10g" \
-rowsetTag "AccessoryPricing" -rowTag "AccessoryDetail" \
"SELECT
   ac.product_id AS AccessoryID,
   ac.name AS Name,
   ac.color AS Color,
   ac.base_price AS BasePrice,
   ac.GetPrice() AS CalculatedPrice,
   ac.supplycontract_ref.supplier_ref.name AS SupplierName
FROM accessories ac"
```

List the details of every supply contract, including the contract ID, cost price, and terms, name and rating of the linked supplier of contracts that expire after year 2025

contract_expire.xml

```
OracleXML getXML -user "grp2/grp22here" -conn
"jdbc:oracle:thin:@sit.itec.yorku.ca:1521/studb10g" \
-dateFormat "yyyy-MM-dd" \
-rowsetTag "ContractExpiries" -rowTag "ContractInfo" \
"SELECT
   sc.supplycontract_id AS ContractID,
   sc.cost_price AS CostPrice,
   sc.expiry_date AS ExpiryDate,
   sc.terms AS Terms,
   sc.supplier_ref.name AS SupplierName,
   sc.supplier_ref.rating AS SupplierRating
FROM supplycontract sc
WHERE sc.expiry_date > DATE '2025-12-31'"
```

12 XQuery Language

```
set long 32000s
Set pagesize 60
```

Find account order id with spending more than $50

```
xquery
let  $c := doc("/public/game4220/game_cartlines.xml")
for $a in $c/VideogameCartLines/CartLine/Order
where $a/OrderTotal > 50
return $a/@orderId
/
```

Find total revenue for PS5 game

```
xquery
let  $c := doc("/public/game4220/platform_sales.xml")
for $v in $c/PlatformSales/Platform
where $v/PlatformName= "PS5"
return $v/TotalRevenue
/
```

Contracts that expire in 2025

```
xquery
let $sc:=doc("/public/game4220/suppliers.xml")
for $c in $sc/Suppliers/Supplier/Contracts/Contract
where starts-with(string($c/ExpiryDate), "2025")
return $c/@contractId
/
```

Find customers who have only one past purchase on their account

```
xquery
let $c := doc("/public/game4220/purchase.xml")
for $ca in $c/CustomerTotals/Customer
where $ca/OrderCount = 1
return $ca/Name/text()
/
```

Find the order made after 2025-09-01

```
xquery
let $c := doc("/public/game4220/customer_orders.xml")
for $o in $c/CustomerAccounts/Account/Order
where xs:date($o/OrderDate) gt xs:date("2025-09-01")
return
<Order>
   <OrderId>{$o/@orderId}</OrderId>
   <OrderDate>{$o/OrderDate/text()}</OrderDate>
</Order>
/
```

Find single order total amount more than $150

```
xquery
let $orders := doc("/public/game4220/order_complete.xml")
for $o in $orders/OrdersWithCartLines/Order
where number($o/OrderInfo/TotalAmount) > 150
return
  <Order orderId="{$o/@orderId}">
    <Date>{$o/OrderInfo/OrderDate/text()}</Date>
```

```
    <Customer>{$o/OrderInfo/Customer/Name/text()}</Customer>
    <TotalAmount>{$o/OrderInfo/TotalAmount/text()}</TotalAmount>
  </Order>
/
```

Finding Registered Customers

```
xquery
let  $c := doc("/public/game4220/customer_accounts.xml")
for $a in $c/CustomerAccounts/Account
where $a/@guestFlag = "N"
return $a
/
```

Find the Supplier with rating C (rating refer to supplier performance and reliability)

```
xquery
let $c := doc("/public/game4220/suppliers.xml")
for $s in $c/Suppliers/Supplier
where normalize-space($s/@rating) = "C"
return
<supplier id="{$s/@supplierId}">
            <name>{$s/Name/text()}</name>
            <rate>{$s/@rating}</rate>
</supplier>
/
```

Find suppliers of games with a base price greater than $60

```
Xquery
let $c := doc("/public/game4220/suppliers.xml")
let $d := doc("/public/game4220/game_supplier.xml")
return
  for $s in $c/Suppliers/Supplier
  for $gs in $d/SupplierVideogames/Supplier
  for $game in $gs/Videogames/Game
  where $s/@supplierId = $gs/@supplierId
    and number($game/BasePrice) > 60
  return
    <Supplier name="{$s/Name/text()}">
      <Game>{$game/Title/text()}</Game>
      <Retail>{$game/BasePrice/text()}</Retail>
    </Supplier>
/
```

Find customer account with cumulative spending above $150

```
xquery
let $purchases := doc("/public/game4220/purchase.xml")
let $accounts := doc("/public/game4220/customer_accounts.xml")
for $c in $purchases/CustomerTotals/Customer
where number($c/TotalSpent) > 150
return
  for $a in $accounts/CustomerAccounts/Account/Customer
  where $a/@customerId = $c/@customerId
  return
    <Customer>
      <Name>{$a/Name/text()}</Name>
      <TotalSpent>{$c/TotalSpent/text()}</TotalSpent>
      <Email>{$a/Email/text()}</Email>
      <Phone>{$a/Phone/text()}</Phone>
    </Customer>
/
```

Find the supplier revenue more than 200

```
xquery
let $revenue := doc("/public/game4220/supplier_revenue.xml")
let $suppliers := doc("/public/game4220/suppliers.xml")
for $r in $revenue/SupplierRevenue/Supplier
where number($r/TotalRevenue) > 200
return
  for $s in $suppliers/Suppliers/Supplier
  where $s/@supplierId = $r/@supplierId
  return
    <Supplier id="{$s/@supplierId}">
      <Name>{$s/Name/text()}</Name>
      <TotalRevenue>{$r/TotalRevenue/text()}</TotalRevenue>
      </Supplier>
/
```

Find game platform sales revenue below $60

```
xquery
let $c := doc("/public/game4220/platform_sales.xml")
let $d := doc("/public/game4220/order_complete.xml")
for $p in $c/PlatformSales/Platform
let $n := $d/OrdersWithCartLines/Order/CartLines/CartLine[Videogame/Platform =
$p/PlatformName]
where number($p/TotalRevenue) < 60
return
  <platform name="{$p/PlatformName/text()}">
```

```
  <revenue>{$p/TotalRevenue/text()}</revenue>
  <orders>{count($n)}</orders>
 </platform>
/
```

XML Documents in the DB

```
SELECT ANY_PATH
  2  FROM RESOURCE_VIEW
  3  WHERE under_path(RES, '/public/game4220') = 1
  4  AND ANY_PATH LIKE '%.xml';

ANY_PATH
--------------------------------------------------
/public/game4220/accessory_cartlines.xml
/public/game4220/accessory_supplier.xml
/public/game4220/account_summary.xml
/public/game4220/contract_expire.xml
/public/game4220/customer_accounts.xml
/public/game4220/customer_orders.xml
/public/game4220/daily_sales.xml
/public/game4220/deal.xml
/public/game4220/game_cartlines.xml
/public/game4220/game_supplier.xml
/public/game4220/order_complete.xml

ANY_PATH
--------------------------------------------------
/public/game4220/override_price.xml
/public/game4220/platform_sales.xml
/public/game4220/purchase.xml
/public/game4220/supplier_revenue.xml
/public/game4220/suppliers.xml

16 rows selected.
```

XSU

```
[freevic@sit ~]$ OracleXML getXML -user "grp2/grp22here" -conn "jdbc:oracle:thin
:@sit.itec.yorku.ca:1521/studb10g" \
-rowsetTag "LowCostGames" -rowTag "GameDeal" \
"SELECT
    vg.product_id AS GameID,
    vg.name AS Title,
    vg.platform AS Platform,
    vg.base_price AS BasePrice,
    vg.supplycontract_ref.cost_price AS ContractCost,
    vg.supplycontract_ref.supplier_ref.name AS Supplier
FROM videogame vg
WHERE vg.supplycontract_ref.cost_price < vg.base_price"
<?xml version = '1.0'?>
<LowCostGames>
   <GameDeal num="1">
      <GAMEID>G0001</GAMEID>
      <TITLE>Cyberpunk 2077</TITLE>
      <PLATFORM>PS5</PLATFORM>
      <BASEPRICE>59.99</BASEPRICE>
      <CONTRACTCOST>50</CONTRACTCOST>
      <SUPPLIER>GameDistributors Inc</SUPPLIER>
   </GameDeal>
   <GameDeal num="2">
      <GAMEID>G0003</GAMEID>
      <TITLE>Flight Simulator</TITLE>
      <PLATFORM>Steam</PLATFORM>
      <BASEPRICE>69.99</BASEPRICE>
      <CONTRACTCOST>40</CONTRACTCOST>
      <SUPPLIER>Digital Games Co</SUPPLIER>
   </GameDeal>
   <GameDeal num="3">
      <GAMEID>G0004</GAMEID>
      <TITLE>Call of Duty</TITLE>
      <PLATFORM>XBOX</PLATFORM>
      <BASEPRICE>69.99</BASEPRICE>
      <CONTRACTCOST>50</CONTRACTCOST>
      <SUPPLIER>GameDistributors Inc</SUPPLIER>
   </GameDeal>
</LowCostGames>
```

```
[freevic@sit ~]$ OracleXML getXML -user "grp2/grp22here" -conn "jdbc:oracle:thin:@sit.itec.yorku.ca:1521/studb10g" \
-rowsetTag "AccountSummary" -rowTag "AccountDetails" \
"SELECT
    a.account_id,
    a.customer_ref.name AS CustomerName,
    a.customer_ref.address AS CustomerAddress,
    (SELECT COUNT(*) FROM accountorder o WHERE o.account_ref = REF(a)) AS TotalOrders
FROM account a"
<?xml version = '1.0'?>
<AccountSummary>
   <AccountDetails num="1">
      <ACCOUNT_ID>A0001</ACCOUNT_ID>
      <CUSTOMERNAME>Johnson Smith</CUSTOMERNAME>
      <CUSTOMERADDRESS>123 Queen St W</CUSTOMERADDRESS>
      <TOTALORDERS>1</TOTALORDERS>
   </AccountDetails>
   <AccountDetails num="2">
      <ACCOUNT_ID>A0002</ACCOUNT_ID>
      <CUSTOMERNAME>Keanu Reeves</CUSTOMERNAME>
      <CUSTOMERADDRESS>156 Hollywood Dr</CUSTOMERADDRESS>
      <TOTALORDERS>1</TOTALORDERS>
   </AccountDetails>
   <AccountDetails num="3">
      <ACCOUNT_ID>A0003</ACCOUNT_ID>
      <CUSTOMERNAME>Lucas Thompson</CUSTOMERNAME>
      <CUSTOMERADDRESS>55 Front St E</CUSTOMERADDRESS>
      <TOTALORDERS>1</TOTALORDERS>
   </AccountDetails>
   <AccountDetails num="4">
      <ACCOUNT_ID>A0004</ACCOUNT_ID>
      <CUSTOMERNAME>Michael Nguyen</CUSTOMERNAME>
      <CUSTOMERADDRESS>940 King St W</CUSTOMERADDRESS>
      <TOTALORDERS>1</TOTALORDERS>
   </AccountDetails>
   <AccountDetails num="5">
      <ACCOUNT_ID>A0005</ACCOUNT_ID>
      <CUSTOMERNAME>Sarah Mitchell</CUSTOMERNAME>
      <CUSTOMERADDRESS>91 Yonge St</CUSTOMERADDRESS>
      <TOTALORDERS>0</TOTALORDERS>
   </AccountDetails>
</AccountSummary>
```

```
[brandl01@sit ~]$ OracleXML getXML -user "grp2/grp22here" -conn "jdbc:oracle:thin:@sit.itec.yorku.ca:1521/studb10g" \
-rowsetTag "AccessoryPricing" -rowTag "AccessoryDetail" \
"SELECT
    ac.product_id AS AccessoryID,
    ac.name AS Name,
    ac.color AS Color,
    ac.base_price AS BasePrice,
    ac.GetPrice() AS CalculatedPrice,
    ac.supplycontract_ref.supplier_ref.name AS SupplierName
FROM accessories ac"
<?xml version = '1.0'?>
<AccessoryPricing>
   <AccessoryDetail num="1">
      <ACCESSORYID>A0001</ACCESSORYID>
      <NAME>HyperX Cloud</NAME>
      <COLOR>black</COLOR>
      <BASEPRICE>129.99</BASEPRICE>
      <CALCULATEDPRICE>129.99</CALCULATEDPRICE>
      <SUPPLIERNAME>Accessory World</SUPPLIERNAME>
   </AccessoryDetail>
   <AccessoryDetail num="2">
      <ACCESSORYID>A0002</ACCESSORYID>
      <NAME>Charge Type C</NAME>
      <COLOR>white</COLOR>
      <BASEPRICE>29.99</BASEPRICE>
      <CALCULATEDPRICE>29.99</CALCULATEDPRICE>
      <SUPPLIERNAME>TechParts Ltd</SUPPLIERNAME>
   </AccessoryDetail>
   <AccessoryDetail num="3">
      <ACCESSORYID>A0003</ACCESSORYID>
      <NAME>PS5 Controller</NAME>
      <COLOR>gold</COLOR>
      <BASEPRICE>79.99</BASEPRICE>
      <CALCULATEDPRICE>89.99</CALCULATEDPRICE>
      <SUPPLIERNAME>Accessory World</SUPPLIERNAME>
   </AccessoryDetail>
   <AccessoryDetail num="4">
      <ACCESSORYID>A0004</ACCESSORYID>
      <NAME>Logitech GPX2</NAME>
      <COLOR>silver</COLOR>
      <BASEPRICE>59.99</BASEPRICE>
      <CALCULATEDPRICE>59.99</CALCULATEDPRICE>
      <SUPPLIERNAME>Accessory World</SUPPLIERNAME>
   </AccessoryDetail>
</AccessoryPricing>
```

```
[brandl01@sit ~]$ OracleXML getXML -user "grp2/grp22here" -conn "jdbc:oracle:thin:@sit.itec.yorku.ca:1521/studb10g" \
-dateFormat "yyyy-MM-dd" \
-rowsetTag "ContractExpiries" -rowTag "ContractInfo" \
"SELECT
    sc.supplycontract_id AS ContractID,
    sc.cost_price AS CostPrice,
    sc.expiry_date AS ExpiryDate,
    sc.terms AS Terms,
    sc.supplier_ref.name AS SupplierName,
    sc.supplier_ref.rating AS SupplierRating
FROM supplycontract sc
WHERE sc.expiry_date > DATE '2025-12-31'"
<?xml version = '1.0'?>
<ContractExpiries>
   <ContractInfo num="1">
      <CONTRACTID>S010</CONTRACTID>
      <COSTPRICE>25</COSTPRICE>
      <EXPIRYDATE>2026-03-15</EXPIRYDATE>
      <TERMS>only gaming accessories</TERMS>
      <SUPPLIERNAME>Accessory World</SUPPLIERNAME>
      <SUPPLIERRATING>A </SUPPLIERRATING>
   </ContractInfo>
   <ContractInfo num="2">
      <CONTRACTID>S029</CONTRACTID>
      <COSTPRICE>40</COSTPRICE>
      <EXPIRYDATE>2028-08-21</EXPIRYDATE>
      <TERMS>send report every three months</TERMS>
      <SUPPLIERNAME>Digital Games Co</SUPPLIERNAME>
      <SUPPLIERRATING>C </SUPPLIERRATING>
   </ContractInfo>
   <ContractInfo num="3">
      <CONTRACTID>S018</CONTRACTID>
      <COSTPRICE>15</COSTPRICE>
      <EXPIRYDATE>2026-08-15</EXPIRYDATE>
      <TERMS>ensure 98% on time delivery</TERMS>
      <SUPPLIERNAME>TechParts Ltd</SUPPLIERNAME>
      <SUPPLIERRATING>B </SUPPLIERRATING>
   </ContractInfo>
</ContractExpiries>
```

XML DOCUMENT

```
SQL> select xmlroot(xmlelement("OrdersWithCartLines",
  2  xmlagg(xmlelement("Order", xmlattributes(o.accountorder_id as "orderId"), xmlelement("Or
derInfo", xmlforest(o.order_date as "OrderDate", o.total_amount as "TotalAmount" ), xmlelemen
t("Customer", xmlattributes(c.customer_id as "customerId"), xmlelement("Name", c.name))), xml
element("CartLines",
  3  (select xmlagg(x) from (
  4  -- videogame lines
  5  select xmlelement("CartLine", xmlattributes(clv.cartline_id as "cartlineId", 'Videogame'
 as "productType"), xmlforest( clv.quantity as "Quantity", clv.unit_price as "UnitPrice"), xm
lelement("Videogame", xmlforest( vg.product_id   as "ProductId", vg.name as "Name", vg.base_p
rice as "BasePrice", vg.platform as "Platform", vg.estb_rating  as "Rating")) ) as x from car
tline clv, videogame vg
where clv.order_ref = ref(o) and clv.product_ref = ref(vg)
  7  union all
  8  -- accessory lines
select xmlelement("CartLine", xmlattributes(cla.cartline_id as "cartlineId", 'Accessory' as "
productType"), xmlforest(cla.quantity as "Quantity",cla.unit_price as "UnitPrice"),
xmlelement("Accessory", xmlforest( ac.product_id as "ProductId", ac.name as "Name", ac.base_p
rice as "BasePrice", ac.accessory_type as "Type",ac.color as "Color" ))) as x
 11  from cartline cla, accessories ac
where cla.order_ref   = ref(o) and cla.product_ref = ref(ac)))))),
 13  version '1.0' ) as document
 14  from accountorder o, account a, customer c
 15  where o.account_ref  = ref(a) and a.customer_ref = ref(c);
```

```xml
<?xml version="1.0"?>
<OrdersWithCartLines>
  <Order orderId="O00001">
    <OrderInfo>
      <OrderDate>2025-10-15</OrderDate>
      <TotalAmount>179.97</TotalAmount>
      <Customer customerId="00001">
        <Name>Johnson Smith</Name>
      </Customer>
    </OrderInfo>
    <CartLines>
      <CartLine cartlineId="CL00001" productType="Videogame">
        <Quantity>1</Quantity>
        <UnitPrice>59.99</UnitPrice>
        <Videogame>
          <ProductId>G0001</ProductId>
          <Name>Cyberpunk 2077</Name>
          <BasePrice>59.99</BasePrice>
          <Platform>PS5</Platform>
          <Rating>M </Rating>
        </Videogame>
      </CartLine>
      <CartLine cartlineId="CL00002" productType="Accessory">
        <Quantity>1</Quantity>
        <UnitPrice>129.99</UnitPrice>
        <Accessory>
          <ProductId>A0001</ProductId>
          <Name>HyperX Cloud</Name>
          <BasePrice>129.99</BasePrice>
          <Type>headset</Type>
          <Color>black</Color>
        </Accessory>
      </CartLine>
    </CartLines>
  </Order>
  <Order orderId="O00002">
    <OrderInfo>
      <OrderDate>2025-10-18</OrderDate>
      <TotalAmount>69.99</TotalAmount>
      <Customer customerId="05011">
        <Name>Michael Nguyen</Name>
      </Customer>
    </OrderInfo>
    <CartLines>
      <CartLine cartlineId="CL00003" productType="Videogame">
```

```
DOCUMENT
--------------------------------------------------------------------------------
  <Order orderId="O00004">
    <OrderInfo>
      <OrderDate>2025-10-22</OrderDate>
      <TotalAmount>189.98</TotalAmount>
      <Customer customerId="00325">
        <Name>Lucas Thompson</Name>
      </Customer>
    </OrderInfo>
    <CartLines>
      <CartLine cartlineId="CL00005" productType="Accessory">
        <Quantity>1</Quantity>
        <UnitPrice>79.99</UnitPrice>
        <Accessory>
          <ProductId>A0003</ProductId>
          <Name>PS5 Controller</Name>
          <BasePrice>79.99</BasePrice>
          <Type>gamepad</Type>
          <Color>gold</Color>
        </Accessory>
      </CartLine>
      <CartLine cartlineId="CL00006" productType="Accessory">
        <Quantity>1</Quantity>
        <UnitPrice>79.99</UnitPrice>
        <Accessory>
          <ProductId>A0003</ProductId>
          <Name>PS5 Controller</Name>
          <BasePrice>79.99</BasePrice>
          <Type>gamepad</Type>
          <Color>gold</Color>
        </Accessory>
      </CartLine>
    </CartLines>
  </Order>
  <Order orderId="O00003">
    <OrderInfo>
```

```
select xmlroot(xmlelement("CustomerTotals",
  2             XMLAGG(XMLELEMENT("Customer",
  3                     XMLATTRIBUTES(t.customer_id AS "customerId"),
 XMLFOREST(t.name AS "Name", t.order_count AS "OrderCount", t.total_spent AS "TotalSpent"))))
,VERSION '1.0') AS document
  5  FROM (SELECT
  6               -- Select Customer attributes
  7               a.customer_ref.customer_id AS customer_id,
  8               a.customer_ref.name AS name,
  9               -- Aggregate Orders
 10                COUNT(o.accountorder_id) AS order_count,
 11                SUM(o.total_amount) AS total_spent
 12     FROM account a, accountorder o
 13        WHERE o.account_ref.account_id = a.account_id
 14        GROUP BY a.customer_ref.customer_id, a.customer_ref.name) t;

DOCUMENT
--------------------------------------------------------------------------------
<?xml version="1.0"?>
<CustomerTotals>
  <Customer customerId="00325">
    <Name>Lucas Thompson</Name>
    <OrderCount>1</OrderCount>
    <TotalSpent>189.98</TotalSpent>
  </Customer>
  <Customer customerId="00001">
    <Name>Johnson Smith</Name>
    <OrderCount>1</OrderCount>
    <TotalSpent>179.97</TotalSpent>
  </Customer>
  <Customer customerId="05011">
    <Name>Michael Nguyen</Name>
    <OrderCount>1</OrderCount>
    <TotalSpent>69.99</TotalSpent>
  </Customer>
  <Customer customerId="00123">
    <Name>Keanu Reeves</Name>
    <OrderCount>1</OrderCount>
    <TotalSpent>129.99</TotalSpent>
  </Customer>
</CustomerTotals>
```

```
select xmlroot(xmlelement("PlatformSales",
xmlagg(xmlelement("Platform", xmlforest( t.platform as "PlatformName", t.total_qty as "TotalQ
uantity", t.total_revenue as "TotalRevenue")))), version '1.0') as document
  3  from (select vg.platform, sum(cl.quantity) as total_qty, sum(cl.quantity * cl.unit_price
) as total_revenue from cartline cl, accountorder o, videogame vg where cl.order_ref  = ref(o
) and cl.product_ref = ref(vg) group by vg.platform) t;

DOCUMENT
--------------------------------------------------------------------------------
<?xml version="1.0"?>
<PlatformSales>
  <Platform>
    <PlatformName>Steam</PlatformName>
    <TotalQuantity>1</TotalQuantity>
    <TotalRevenue>69.99</TotalRevenue>
  </Platform>
  <Platform>
    <PlatformName>XBOX</PlatformName>
    <TotalQuantity>1</TotalQuantity>
    <TotalRevenue>69.99</TotalRevenue>
  </Platform>
  <Platform>
    <PlatformName>PS5</PlatformName>
    <TotalQuantity>1</TotalQuantity>
    <TotalRevenue>59.99</TotalRevenue>
  </Platform>
</PlatformSales>
```

```
SQL> SELECT XMLROOT(XMLELEMENT("DailySales",
  2              XMLAGG(
  3                  XMLELEMENT("Day", XMLFOREST(
  4                          t.order_date AS "OrderDate",
  5                          t.order_count AS "OrderCount",
                        t.daily_total AS "TotalAmount")
                    ))), VERSION '1.0') AS document
  8  FROM (SELECT
  9                   o.order_date,
 10                   COUNT(o.accountorder_id) AS order_count,
 11                   SUM(o.total_amount) AS daily_total
 12       FROM accountorder o
 13    GROUP BY o.order_date) t;

DOCUMENT
--------------------------------------------------------------------------------
<?xml version="1.0"?>
<DailySales>
  <Day>
    <OrderDate>2025-10-20</OrderDate>
    <OrderCount>1</OrderCount>
    <TotalAmount>129.99</TotalAmount>
  </Day>
  <Day>
    <OrderDate>2025-10-18</OrderDate>
    <OrderCount>1</OrderCount>
    <TotalAmount>69.99</TotalAmount>
  </Day>
  <Day>
    <OrderDate>2025-10-15</OrderDate>
    <OrderCount>1</OrderCount>
    <TotalAmount>179.97</TotalAmount>
  </Day>
  <Day>
    <OrderDate>2025-10-22</OrderDate>
    <OrderCount>1</OrderCount>
    <TotalAmount>189.98</TotalAmount>
  </Day>
</DailySales>
```

```
select xmlroot(xmlelement("SupplierRevenue", xmlagg(xmlelement("Supplier", xmlattributes(t.s_
id as "supplierId"), xmlforest( t.s_name as "Name", t.total_revenue as "TotalRevenue")))), ve
rsion '1.0' ) as document
  2  from (select s_id, s_name, sum(amount) as total_revenue from (
  3  -- revenue from videogames
  4  select s.supplier_id as s_id, s.name as s_name, cl.quantity * cl.unit_price as amount
  5  from supplier s, supplycontract sc, videogame vg, cartline cl, accountorder o
where sc.supplier_ref= ref(s)and vg.supplycontract_ref = ref(sc) and cl.product_ref= ref(vg)
and cl.order_ref= ref(o)
  7  union all
  8  -- revenue from accessories
  9  select s.supplier_id as s_id, s.name as s_name, cl.quantity * cl.unit_price as amount
 10  from supplier s, supplycontract sc, accessories ac, cartline cl, accountorder o
 11  where sc.supplier_ref = ref(s)  and ac.supplycontract_ref = ref(sc) and cl.product_ref =
 ref(ac) and cl.order_ref = ref(o) ) group by s_id, s_name) t;

DOCUMENT
--------------------------------------------------------------------------------
<?xml version="1.0"?>
<SupplierRevenue>
  <Supplier supplierId="S003">
    <Name>Accessory World</Name>
    <TotalRevenue>289.97</TotalRevenue>
  </Supplier>
  <Supplier supplierId="S001">
    <Name>GameDistributors Inc</Name>
    <TotalRevenue>129.98</TotalRevenue>
  </Supplier>
  <Supplier supplierId="S004">
    <Name>Digital Games Co</Name>
    <TotalRevenue>69.99</TotalRevenue>
  </Supplier>
</SupplierRevenue>
```

XQUERY

```
xquery
  2  let  $c := doc("/public/game4220/game_cartlines.xml")
  3  for $a in $c/VideogameCartLines/CartLine/Order
  4  where $a/OrderTotal > 50
  5  return $a/@orderId
  6  /

Result Sequence
--------------------------------------------------------------------------------
O00001
O00002
O00003
```

```
xquery
  2  let  $c := doc("/public/game4220/platform_sales.xml")
  3  for $v in $c/PlatformSales/Platform
  4  where $v/PlatformName= "PS5"
  5  return $v/TotalRevenue
  6  /

Result Sequence
--------------------------------------------------------------------------------
<TotalRevenue>59.99</TotalRevenue>
```

```
xquery
  2  let $sc:=doc("/public/game4220/suppliers.xml")
  3  for $c in $sc/Suppliers/Supplier/Contracts/Contract
where starts-with(string($c/ExpiryDate), "2025")
  5  return $c/@contractId
  6  /

Result Sequence
------------------------------------------------------------
S026
```

```
xquery
  2  let $c := doc("/public/game4220/purchase.xml")
  3  for $ca in $c/CustomerTotals/Customer
  4  where $ca/OrderCount = 1
  5  return $ca/Name/text()
  6  /

Result Sequence
------------------------------------------------------------
Lucas Thompson
Johnson Smith
Michael Nguyen
Keanu Reeves
```

```
xquery
  2  let $c := doc("/public/game4220/customer_orders.xml")
  3  for $o in $c/CustomerAccounts/Account/Order
where xs:date($o/OrderDate) gt xs:date("2025-09-01")
  5  return
  6  <Order>
  7     <OrderId>{$o/@orderId}</OrderId>
   <OrderDate>{$o/OrderDate/text()}</OrderDate>
  9  </Order>
 10  /

Result Sequence
------------------------------------------------------------------------------
<Order><OrderId orderId="O00001"></OrderId><OrderDate>2025-10-15</OrderDate></Or
<Order><OrderId orderId="O00002"></OrderId><OrderDate>2025-10-18</OrderDate></Or
<Order><OrderId orderId="O00004"></OrderId><OrderDate>2025-10-22</OrderDate></Or
<Order><OrderId orderId="O00003"></OrderId><OrderDate>2025-10-20</OrderDate></Or
```

```
xquery
let $orders := doc("/public/game4220/order_complete.xml")
  2     3  for $o in $orders/OrdersWithCartLines/Order
  4   where number($o/OrderInfo/TotalAmount) > 150
  5   return
  6     <Order orderId="{$o/@orderId}">
      <Date>{$o/OrderInfo/OrderDate/text()}</Date>
      <Customer>{$o/OrderInfo/Customer/Name/text()}</Customer>
      <TotalAmount>{$o/OrderInfo/TotalAmount/text()}</TotalAmount>
 10     </Order>
 11   /

Result Sequence
--------------------------------------------------------------------------------
<Order orderId="O00001"><Date>2025-10-15</Date><Customer>Johnson Smith</Customer
<Order orderId="O00004"><Date>2025-10-22</Date><Customer>Lucas Thompson</Custome
```

Video Game store System

ITEC 4220

Wing Yan Kwan - 219474477

Victor Kwan - 216436990

Feng-Yuan Lee - 218501643

Brandon Lu - 218501254

Intro + 12 XML Documents:

1. **List all accounts with their linked customers, showing account ID, guest flag, and customer name, address, email, and phone. -** *(customer_accounts.xml)*
2. **For every account ID, show the customer information, order ID, date, and total amount. -***(customer_orders.xml)*
3. **For cart lines that contain videogames, show the order, customer, and videogame information** *-(game_cartlines.xml)*
4. **For all cart lines that contain accessories, show the order, customer, accessory information** *-(accessory_cartlines.xml)*
5. **List every supplier information and contracts including contract ID, terms , price, expiry date, and supplier rating.** *-(suppliers.xml)*
6. **For every supplier, list all videogames they supply, including product ID, name, price, platform, rating, and contract price.** *-(game_supplier.xml)*

12 XML Documents:

7. **For every supplier, list all accessories they supply, including product ID, name, price, type, colour, and contract price.** - *(accessory_supplier.xml)*

8. **For each customer, compute and show the total amount they have spent and the number of orders they placed.** - *(order_complete.xml)*

9. **For each customer, show how many purchases their account made (Group By)** -*(purchase.xml)*

10. **For each videogame platform (e.g., PS5, Switch, XBOX, Steam), show total quantity sold and total revenue. (Group By)** - *(platform_sales.xml)*

11. **Show daily sales revenue generated (Group By)** - *(daily_sales.xml)*

12. **For each supplier, show the total revenue generated from all videogames and accessories supplied by them (Group By)** - *(supplier_revenue.xml)*

Oracle XSU:

1. List all videogames where the contract cost is less than the public base price. Show the game ID, title, platform, base price, and the exact contract cost and supplier name.

2. List all accounts, including the customer's name and address, and include the total number of orders placed by that specific account.

3. List all accessories information and the calculated price using the override GetPrice() method. Include the name of the supplier responsible for the contract.

4. List the details of every supply contract, including the contract ID, cost price, and terms, name and rating of the linked supplier of contracts that expire after year 2025

XML Documents:

1. ***List all accounts with their linked customers, showing account ID, guest flag, and customer name, address, email, and phone:***

Accounts and their customers

```
select xmlroot(xmlelement("CustomerAccounts",
    xmlagg(
        xmlelement("Account", xmlattributes(a.account_id AS "accountId",
        a.guestLogin AS "guestFlag"),
            xmlelement("Customer", xmlattributes(
            a.customer_ref.customer_id AS "customerId")
            xmlforest(
              a.customer_ref.name AS "Name",
              a.customer_ref.address AS "Address",
              a.customer_ref.email AS "Email",
              a.customer_ref.phoneNum AS "Phone"))))),
    VERSION '1.0') AS document from account a;
```

2. ***For every account ID, show the customer information, order ID, date, and total amount:***

Orders with account and customer details

```
select xmlroot(xmlelement("CustomerAccounts",
      xmlagg(
        xmlelement("Account", xmlattributes(a.account_id AS "accountId"),
          xmlelement("Customer",
            xmlattributes(a.customer_ref.customer_id AS "customerId"),
            xmlforest(
              a.customer_ref.name AS "Name",
              a.customer_ref.address AS "Address",
              a.customer_ref.email AS "Email",
              a.customer_ref.phoneNum AS "Phone")),
          (select xmlagg(
              xmlelement("Order",
                xmlattributes(o.accountorder_id AS "orderId"),
                xmlforest(
                  o.order_date AS "OrderDate",
                  o.total_amount AS "TotalAmount")))
            from accountorder o
            where o.account_ref = REF(a))
))), VERSION '1.0') AS document from account a;
```

3. ***For cart lines that contain videogames, show the order, customer, and videogame information:***

Videogame cart lines per order

```
select xmlroot(xmlelement("VideogameCartLines",
xmlagg( xmlelement("CartLine", xmlattributes(cl.cartline_id  as "cartlineId",  cl.quantity
as "Quantity"),
xmlelement("Order", xmlattributes(o.accountorder_id as "orderId", o.order_date  as
"OrderDate"), xmlelement("OrderTotal", o.total_amount),
xmlelement("Customer", xmlattributes(c.customer_id as "customerId"), xmlelement("Name",
c.name))),
xmlelement("Videogame", xmlattributes( vg.product_id AS "GameId", vg.platform AS
"Platform", vg.estb_rating AS "Rating"), xmlforest(vg.name as "Title", vg.base_price as
"BasePrice"), xmlelement("UnitPriceCharged", cl.unit_price)))))), version '1.0') as document
from cartline  cl, accountorder o, account  a, customer  c, videogame vg
where cl.order_ref = ref(o) and o.account_ref  = ref(a) and a.customer_ref = ref(c)  and
cl.product_ref = ref(vg);
```

4. ***For all cart lines that contain accessories, show the order, customer, accessory information:***

Accessory cart lines per order

```
select xmlroot(xmlelement("AccessoryCartLines",
xmlagg( xmlelement("CartLine", xmlattributes(cl.cartline_id  as "cartlineId",  cl.quantity  as
"Quantity"),
xmlelement("Order", xmlattributes(o.accountorder_id as "orderId", o.order_date  as
"OrderDate"), xmlelement("OrderTotal", o.total_amount),
xmlelement("Customer", xmlattributes(c.customer_id as "customerId"), xmlelement("Name",
c.name))),
 xmlelement("Accessory", xmlattributes(ac.product_id as "AccessoryId", ac.accessory_type
as "Type", ac.color  as "Color"),
xmlforest(ac.name as "Name", ac.base_price as "BasePrice"),
xmlelement("UnitPriceCharged", cl.unit_price)))))),
version '1.0') as document from cartline  cl, accountorder o, account  a, customer  c,
accessories  ac
where cl.order_ref = ref(o) and o.account_ref  = ref(a)  and a.customer_ref = ref(c) and
cl.product_ref = ref(ac);
```

5. ***List every supplier information and contracts including contract ID, terms , price, expiry date, and supplier rating:***

Suppliers and their supply contracts

```
select xmlroot(xmlelement("Suppliers",
        xmlagg(
          xmlelement("Supplier",
            xmlattributes(s.supplier_id AS "supplierId", s.rating AS "rating"),
        xmlforest(s.name AS "Name", s.address AS "Address", s.phone AS  "Phone"),
            xmlelement("Contracts",
              (select xmlagg(
                      xmlelement("Contract",
                        xmlattributes(sc.supplycontract_id AS "contractId"),
                         xmlforest(sc.terms AS "Terms", sc.cost_price AS "CostPrice",
                      sc.expiry_date AS "ExpiryDate")))
                 from supplycontract sc
                 where sc.supplier_ref.supplier_id = s.supplier_id)
    )))), VERSION '1.0') AS document from supplier s;
```

6. ***For every supplier, list all videogames they supply, including product ID, name, price, platform, rating, and contract price:***

Suppliers and their videogames

```
select xmlroot(xmlelement("SupplierVideogames",
        xmlagg(
          xmlelement("Supplier",
            xmlattributes(s.supplier_id AS "supplierId"),
            xmlelement("Name", s.name),
            xmlelement("Videogames",
              (select xmlagg(
                      xmlelement("Game",
                        xmlattributes(vg.product_id AS "gameId"),
                        xmlforest(
                          vg.name AS "Title",
                          vg.base_price AS "BasePrice",
                          vg.platform AS "Platform",
                          vg.estb_rating AS "Rating",
                          sc.cost_price AS "ContractCost")))
                 from supplycontract sc, videogame vg
                 where sc.supplier_ref.supplier_id = s.supplier_id AND
               vg.supplycontract_ref.supplycontract_id = sc.supplycontract_id)
    )))), VERSION '1.0') AS document from supplier s;
```

7. ***For every supplier, list all accessories they supply, including product ID, name, price, type, colour, and contract price:***

Suppliers and their accessories

```
select xmlroot(xmlelement("SupplierAccessories",
xmlagg( xmlelement("Supplier", xmlattributes(s.supplier_id as "supplierId"),
xmlelement("Name", s.name), xmlelement("Accessories", (select xmlagg(
xmlelement("Accessory", xmlattributes(ac.product_id as "accessoryId"),
xmlforest( ac.name as "Name", ac.base_price as "BasePrice", ac.accessory_type as "Type",
ac.color as "Color", sc.cost_price as "ContractCost" )))
from supplycontract sc, accessories ac
where sc.supplier_ref = ref(s) and ac.supplycontract_ref = ref(sc))))),
version '1.0') as document from supplier s;
```

8. ***For each customer, compute and show the total amount they have spent and the number of orders they placed:***

Orders with all cart lines (videogames + accessories)

```
select xmlroot(xmlelement("OrdersWithCartLines",
xmlagg(xmlelement("Order", xmlattributes(o.accountorder_id as "orderId"),
xmlelement("OrderInfo", xmlforest(o.order_date as "OrderDate", o.total_amount as
"TotalAmount" ), xmlelement("Customer", xmlattributes(c.customer_id as "customerId"),
xmlelement("Name", c.name))), xmlelement("CartLines",
(select xmlagg(x) from (
-- videogame lines
select xmlelement("CartLine", xmlattributes(clv.cartline_id as "cartlineId", 'Videogame' as
"productType"), xmlforest( clv.quantity as "Quantity", clv.unit_price as "UnitPrice"),
xmlelement("Videogame", xmlforest( vg.product_id  as "ProductId", vg.name as "Name",
vg.base_price as "BasePrice", vg.platform as "Platform", vg.estb_rating  as "Rating")) ) as x
from cartline clv, videogame vg
where clv.order_ref = ref(o) and clv.product_ref = ref(vg)
union all
-- accessory lines
select xmlelement("CartLine", xmlattributes(cla.cartline_id as "cartlineId", 'Accessory' as
"productType"), xmlforest(cla.quantity as "Quantity",cla.unit_price as "UnitPrice"),
xmlelement("Accessory", xmlforest( ac.product_id as "ProductId", ac.name as "Name",
ac.base_price as "BasePrice", ac.accessory_type as "Type",ac.color as "Color" ))) as x
from cartline cla, accessories ac
where cla.order_ref  = ref(o) and cla.product_ref = ref(ac))))))),
version '1.0' ) as document
from accountorder o, account a, customer c
where o.account_ref  = ref(a) and a.customer_ref = ref(c);
```

9. ***For each customer, show how many purchases their account made (Group By):***

number of orders per customer and total spent

```
select xmlroot(xmlelement("CustomerTotals",
        XMLAGG(XMLELEMENT("Customer",
            XMLATTRIBUTES(t.customer_id AS "customerId"),
                XMLFOREST(t.name AS "Name", t.order_count AS "OrderCount",
                t.total_spent AS "TotalSpent")))),VERSION '1.0') AS document
FROM (SELECT
        -- Select Customer attributes
         a.customer_ref.customer_id AS customer_id,
         a.customer_ref.name AS name,
         -- Aggregate Orders
          COUNT(o.accountorder_id) AS order_count,
          SUM(o.total_amount) AS total_spent
          FROM account a, accountorder o
          WHERE o.account_ref.account_id = a.account_id
          GROUP BY a.customer_ref.customer_id, a.customer_ref.name) t;
```

10. ***For each videogame platform (e.g., PS5, Switch, XBOX, Steam), show total quantity sold and total revenue (Group By):***

Sales per videogame platform

```
select xmlroot(xmlelement("PlatformSales",
xmlagg(xmlelement("Platform", xmlforest( t.platform as "PlatformName", t.total_qty as
"TotalQuantity", t.total_revenue as "TotalRevenue"))), version '1.0') as document
from (select vg.platform, sum(cl.quantity) as total_qty, sum(cl.quantity * cl.unit_price) as
total_revenue from cartline cl, accountorder o, videogame vg where cl.order_ref  = ref(o) and
cl.product_ref = ref(vg) group by vg.platform) t;
```

11. ***Show daily sales revenue generated (Group By):***

Daily sales per order date

```
select xmlroot(xmlelement("DailySales",
xmlagg(xmlelement("Day", xmlforest(t.order_date   as "OrderDate", t.order_count  as
"OrderCount", t.daily_total as "TotalAmount" ) )) ),
version '1.0') as document
from (select o.order_date, count(o.accountorder_id) as order_count, sum(o.total_amount) as
daily_total from accountorder o, account a where o.account_ref = ref(a) group by
o.order_date) t;
```

12. ***For each supplier, show the total revenue generated from all videogames and accessories supplied by them (Group By):***

Total revenue per supplier (games + accessories)

```
select xmlroot(xmlelement("SupplierRevenue", xmlagg(xmlelement("Supplier",
xmlattributes(t.s_id as "supplierId"), xmlforest( t.s_name as "Name", t.total_revenue as
"TotalRevenue")))), version '1.0' ) as document
from (select s_id, s_name, sum(amount) as total_revenue from (
-- revenue from videogames
select s.supplier_id as s_id, s.name as s_name, cl.quantity * cl.unit_price as amount
from supplier s, supplycontract sc, videogame vg, cartline cl, accountorder o
where sc.supplier_ref= ref(s)and vg.supplycontract_ref = ref(sc) and cl.product_ref= ref(vg)
and cl.order_ref= ref(o)
union all
-- revenue from accessories
select s.supplier_id as s_id, s.name as s_name, cl.quantity * cl.unit_price as amount
from supplier s, supplycontract sc, accessories ac, cartline cl, accountorder o
where sc.supplier_ref = ref(s)  and ac.supplycontract_ref = ref(sc) and cl.product_ref =
ref(ac) and cl.order_ref = ref(o) ) group by s_id, s_name) t;
```

Oracle XSU(four):

1. **List all videogames where the contract cost is less than the public base price. Show the game ID, title, platform, base price, and the exact contract cost and supplier name.** (***deal.xml***)

```
OracleXML getXML -user "grp2/grp22here" -conn
"jdbc:oracle:thin:@sit.itec.yorku.ca:1521/studb10g" \
-rowsetTag "LowCostGames" -rowTag "GameDeal" \
"SELECT
   vg.product_id AS GameID,
   vg.name AS Title,
   vg.platform AS Platform,
   vg.base_price AS BasePrice,
   vg.supplycontract_ref.cost_price AS ContractCost,
   vg.supplycontract_ref.supplier_ref.name AS Supplier
FROM videogame vg
WHERE vg.supplycontract_ref.cost_price < vg.base_price"
```

2. **List all accounts, including the customer's name and address, and include the total number of orders placed by that specific account.** (***account_summary.xml***)

```
OracleXML getXML -user "grp2/grp22here" -conn
"jdbc:oracle:thin:@sit.itec.yorku.ca:1521/studb10g" \
-rowsetTag "AccountSummary" -rowTag "AccountDetails" \
"SELECT
   a.account_id,
   a.customer_ref.name AS CustomerName,
   a.customer_ref.address AS CustomerAddress,
   (SELECT COUNT(*) FROM accountorder o WHERE o.account_ref = REF(a)) AS
TotalOrders
FROM account a"
```

3. **List all accessories information and the calculated price using the override GetPrice() method. Include the name of the supplier responsible for the contract. (*override_price.xml*)**

```
OracleXML getXML -user "grp2/grp22here" -conn
"jdbc:oracle:thin:@sit.itec.yorku.ca:1521/studb10g" \
-rowsetTag "AccessoryPricing" -rowTag "AccessoryDetail" \
"SELECT
   ac.product_id AS AccessoryID,
   ac.name AS Name,
   ac.color AS Color,
   ac.base_price AS BasePrice,
   ac.GetPrice() AS CalculatedPrice,
   ac.supplycontract_ref.supplier_ref.name AS SupplierName
FROM accessories ac"
```

4. **List the details of every supply contract, including the contract ID, cost price, and terms, name and rating of the linked supplier of contracts that expire after year 2025. (*contract_expire.xml*)**

```
OracleXML getXML -user "grp2/grp22here" -conn
"jdbc:oracle:thin:@sit.itec.yorku.ca:1521/studb10g" \
-dateFormat "yyyy-MM-dd" \
-rowsetTag "ContractExpiries" -rowTag "ContractInfo" \
"SELECT
   sc.supplycontract_id AS ContractID,
   sc.cost_price AS CostPrice,
   sc.expiry_date AS ExpiryDate,
   sc.terms AS Terms,
   sc.supplier_ref.name AS SupplierName,
   sc.supplier_ref.rating AS SupplierRating
FROM supplycontract sc
WHERE sc.expiry_date > DATE '2025-12-31'"
```

12 XML Queries:

1. ***Find account order id with spending more than $50:***

```
xquery
let  $c := doc("/public/game4220/game_cartlines.xml")
for $a in $c/VideogameCartLines/CartLine/Order
where $a/OrderTotal > 50
return $a/@orderId
/
```

2. ***Find total revenue for PS5 game:***

```
xquery
let  $c := doc("/public/game4220/platform_sales.xml")
for $v in $c/PlatformSales/Platform
where $v/PlatformName= "PS5"
return $v/TotalRevenue
/
```

12 XML Queries:

3. ***Contracts that expire in 2025:***

```
xquery
  let $sc:=doc("/public/game4220/supplier_contracts.xml")
  for $c in $sc/Suppliers/Supplier/Contracts/Contract
  where starts-with(string($c/ExpiryDate), "2025")
  return $c/@contractId
/
```

4. ***Find customers who have only have one purchase on their account:***

```
xquery
let $c := doc("/public/game4220/purchase.xml")
for $ca in $c/CustomerTotals/Customer
where $ca/OrderCount = 1
return $ca/Name/text()
/
```

12 XML Queries:

5. ***Find the order made after 2025-09-01:***

```
xquery
let $c := doc("/public/game4220/customer_orders.xml")
for $o in $c/CustomerAccounts/Account/Order
where xs:date($o/OrderDate) gt xs:date("2025-09-01")
return
<Order>
  <OrderId>{$o/@orderId}</OrderId>
  <OrderDate>{$o/OrderDate/text()}</OrderDate>
</Order>
/
```

6. ***Find single order total amount more than $150:***

```
xquery
let $orders := doc("/public/game4220/order_complete.xml")
for $o in $orders/OrdersWithCartLines/Order
where number($o/OrderInfo/TotalAmount) > 150
return
  <Order orderId="{$o/@orderId}">
    <Date>{$o/OrderInfo/OrderDate/text()}</Date>
    <Customer>{$o/OrderInfo/Customer/Name/text()}</Customer>
    <TotalAmount>{$o/OrderInfo/TotalAmount/text()}</TotalAmount>
  </Order>
/
```

12 XML Queries:

7. ***Finding Registered Customers:***

```
xquery
let  $c := doc("/public/game4220/customer_accounts.xml")
for $a in $c/CustomerAccounts/Account
where $a/@guestFlag = "N"
return $a
/
```

8. ***Find the Supplier with rating C (rating refer to supplier performance and reliability):***

```
xquery
let $c := doc("/public/game4220/suppliers.xml")
for $s in $c/Suppliers/Supplier
where normalize-space($s/@rating) = "C"
return
<supplier id="{$s/@supplierId}">
        <name>{$s/Name/text()}</name>
        <rate>{$s/@rating}</rate>
</supplier>
/
```

12 XML Queries:

9. ***Find suppliers of games with a base price greater than $60:***

```
Xquery
let $c := doc("/public/game4220/suppliers.xml")
let $d := doc("/public/game4220/game_supplier.xml")
return
 for $s in $c/Suppliers/Supplier
 for $gs in $d/SupplierVideogames/Supplier
 for $game in $gs/Videogames/Game
 where $s/@supplierId = $gs/@supplierId
  and number($game/BasePrice) > 60
 return
  <Supplier name="{$s/Name/text()}">
   <Game>{$game/Title/text()}</Game>
   <Retail>{$game/BasePrice/text()}</Retail>
  </Supplier>
/
```

12 XML Queries:

10. ***Find customer account with cumulative spending above $150:***

```
xquery
let $purchases := doc("/public/game4220/purchase.xml")
let $accounts := doc("/public/game4220/customer_accounts.xml")
for $c in $purchases/CustomerTotals/Customer
where number($c/TotalSpent) > 150
return
  for $a in $accounts/CustomerAccounts/Account/Customer
  where $a/@customerId = $c/@customerId
  return
    <Customer>
      <Name>{$a/Name/text()}</Name>
      <TotalSpent>{$c/TotalSpent/text()}</TotalSpent>
      <Email>{$a/Email/text()}</Email>
      <Phone>{$a/Phone/text()}</Phone>
    </Customer>
/
```

12 XML Queries:

11. ***Find the supplier revenue more than 200:***

```
xquery
let $revenue := doc("/public/game4220/supplier_revenue.xml")
let $suppliers := doc("/public/game4220/suppliers.xml")
for $r in $revenue/SupplierRevenue/Supplier
where number($r/TotalRevenue) > 200
return
  for $s in $suppliers/Suppliers/Supplier
  where $s/@supplierId = $r/@supplierId
  return
    <Supplier id="{$s/@supplierId}">
      <Name>{$s/Name/text()}</Name>
      <TotalRevenue>{$r/TotalRevenue/text()}</TotalRevenue>
    </Supplier>
/
```

12. ***Find game platform sales revenue below $60:***

```
xquery
let $c := doc("/public/game4220/platform_sales.xml")
let $d := doc("/public/game4220/order_complete.xml")
for $p in $c/PlatformSales/Platform
let $n := $d/OrdersWithCartLines/Order/CartLines/CartLine[Videogame/Platform =
$p/PlatformName]
where number($p/TotalRevenue) < 60
return
  <platform name="{$p/PlatformName/text()}">
    <revenue>{$p/TotalRevenue/text()}</revenue>
    <orders>{count($n)}</orders>
  </platform>
/
```